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SUBJECT CODE NO: - RNP-8058
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A) (NEP) Sem - II
Examination March / April 2026
Basics of Civil Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from question number 2, 3, 4, 5, 6 and 7

Q.1 Attempt any 10 of the following.**10x2=20**

1. Cement mortar consists of:
 - a) Cement and steel
 - b) Cement and sand
 - c) Cement and bricks
 - d) Cement and aggregate
2. Foundation transfers building load to:
 - a) Roof
 - b) Soil
 - c) Beam
 - d) Slab
3. A lintel is provided over:
 - a) Roof
 - b) Foundation
 - c) Door and window openings
 - d) Floor
4. Flooring materials include:
 - a) Marble
 - b) Tiles
 - c) Cement concrete
 - d) All of the above
5. Reduced level represents:
 - a) Length of line
 - b) Elevation of point
 - c) Width of road
 - d) Thickness of slab
6. What is pile foundation?
7. What is brick masonry?
8. Define flat RCC roof.
9. What is stretcher bond?
10. Define chain in surveying.
11. State any two components of floor.
12. What is reduced level?
13. Define benchmark in leveling.

Solve any FOUR of the following.**Q.2 a) Explain the properties and uses of****05**

- i) Bricks
- ii) Rubble

b) Explain the following types of foundations with neat sketches:**05**

- i) Cantilever footing
- ii) Pile foundation

- Q.3** a) Explain brick masonry and compare English bond with Flemish bond. **05**
b) Explain the types of lintels used in building construction. **05**
- Q.4** a) Explain header bond and stretcher bond with neat sketches. **05**
b) Explain the following: **05**
i) Casement window
ii) Battened, ledged and braced door
- Q.5** a) Explain trussed roofs with neat sketches **05**
b) Explain the use of prismatic compass and angular measurements in surveying. **05**
- Q.6** a) Explain chain surveying and measurement of distances using chain and tape. **05**
b) Describe flat RCC roof and components of floor construction. **05**
- Q.7** a) Explain the use of dumpy level in leveling operations. **05**
b) Explain the properties and uses of the coarse aggregate. **05**